

Tonmoy Day Sarkar

Software Engineer

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[Linkedin](#) | [GitHub](#) | [Google Scholar](#)

ABOUT ME

Software Engineering graduate (Jain Deemed-to-be University, 2024) with a strong foundation in end-to-end software development and applied machine learning. Demonstrated research practice in predictive analytics, recommendation systems, and data processing, backed by an undergraduate thesis and four peer-reviewed IEEE publications. Skilled in combining full-stack software development with core ML techniques—including collaborative filtering, ensemble methods, geospatial clustering, and hyperparameter tuning. Seeking to leverage hands-on development expertise and research capability to build robust, scalable AI systems and data-driven software solutions.

EDUCATION

Bachelor of Technology — Computer Engineering (Software Engineering) Jul 2020 – Nov 2024

Jain (Deemed-to-be University), Bangalore, India | Final grade: 8.727/10 (EQF Level 6)

- Thesis: E-commerce-based Smart Recommendation System using element-by-element collaborative filtering combined with machine learning techniques.
- Relevant coursework: Machine Learning, Data Structures & Algorithms, Computer Architecture, Theory of Computation, Database Management Systems, Cloud Computing, Information Security, Agile Software Development.

Higher Secondary Certificate (Science) 2018

Govt. Maulana Mohammad Ali College, Tangail, Bangladesh | Final grade: 4.08/5

Secondary School Certificate (Science) 2016

Sristy College of Tangail, Tangail, Bangladesh | Final grade: 5.00/5

RESEARCH EXPERIENCE

Student Research Assistant Sep 2023 – Jul 2024

Supervisor: Mr. Karthikeyan S., Assistant Professor — Jain (Deemed-to-be University), Bangalore, India

- Conducted independent and supervised research resulting in four IEEE conference publications spanning recommendation systems, churn prediction, regression modeling, and geospatial clustering.
- Applied machine learning pipelines end-to-end: data preprocessing, feature engineering, model selection, hyperparameter tuning, and evaluation.

Data Analyst Intern — Brand & Visual Dec 2025 – May 2026

Dhaka, Bangladesh

- Performed data analysis and visualization to support business decision-making, applying statistical and analytical methods to real datasets.

PUBLICATIONS

- Sarkar, T. D. et al. "E-commerce-based Smart Recommendation System Using Element-by-Element Collaborative Filtering with Machine Learning Technology." 2024 International Conference on Communication, Computing and Internet of Things (IC3IoT), IEEE, 2024.
- Sarkar, T. D. et al. "Analysis of Customer Churn for Telecom Company with SMOTE-ENN and Hyperparameter Tuning Randomized-SearchCV Technique in Advanced Machine Learning Technology." 2024 International Conference on Advancements in Power, Communication and Intelligent Systems, IEEE, 2024.
- Sarkar, T. D. et al. "Predicting Restaurant Ratings: A Comparative Study of Linear Regression and Decision Tree Regression Models in Machine Learning Technology." 4th International Conference on Advancement in Electronics & Communication Engineering (AECE), IEEE, 2025.

- Sarkar, T. D. et al. "Comprehensive Geospatial and Statistical Analysis of Restaurant Distribution Using K-means Clustering in Machine Learning Technology." 4th International Conference on Advancement in Electronics & Communication Engineering (AECE), IEEE, 2025.

RESEARCH INTERESTS

Machine Learning • Natural Language Processing • Artificial Intelligence • Neural Networks • Algorithms • Computer Architecture • Applied AI Systems

TECHNICAL SKILLS

- Programming & ML: Python, C/C++, scikit-learn workflows (regression, clustering, ensemble methods, SMOTE-ENN), data analysis
- Web & Databases: JavaScript, Node.js, Express.js, React.js, Next.js, MongoDB, Firebase, HTML/CSS/Bootstrap
- Tools: Microsoft Word, Hadoop (data ingestion), Git/GitHub

CERTIFICATIONS & TRAINING

- Machine Learning Virtual Internship — CodSoft, IT Service
- Certificate of Reviewing — European Journal of Computer Sciences and Informatics
- Data Analysis Using Python — IBM (Cognitive Class)
- Asia AI Odyssey Challenge — Microsoft
- Moving Data into Hadoop (BD0131EN) — Big Data University
- Data Analytics and Visualization — Accenture North America (Forage)

HONOURS & AWARDS

- Study in India (G2) Scholarship — Jain (Deemed-to-be University), 2021
- The Chancellor's Scholarship — Jain (Deemed-to-be University), 2020

LANGUAGE SKILLS

Bengali (native) | English

REFERENCES

- Dr. Arvind K. S., Associate Professor, Jain (Deemed-to-be University), India — (+91) 9944323386, ks.arvind@jainuniversity.ac.in
- Mr. Karthikeyan S., Assistant Professor, Jain (Deemed-to-be University), India — (+91) 9543277017, s.karthikeyan@jainuniversity.ac.in